Covariance

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Covariance provides a measure of the strength of the correlation between two or more sets of random variates. The covariance for two random variates X and Y, each with sample size N, is defined by the expectation value

$$\operatorname{cov}(X, Y) = \langle (X - \mu_X)(Y - \mu_Y) \rangle \tag{1}$$

$$= \langle X\, Y \rangle - \mu_X\, \mu_y \tag{2}$$

where $\mu_x = \langle X \rangle$ and $\mu_y = \langle Y \rangle$ are the respective means, which can be written out explicitly as

$$\operatorname{cov}(X, Y) = \sum_{i=1}^{N} \frac{(x_i - \bar{x})(y_i - \bar{y})}{N}. \tag{3}$$

For uncorrelated variates,

$$\operatorname{cov}(X, Y) = \langle X\, Y \rangle - \mu_X\, \mu_Y = \langle X \rangle \langle Y \rangle - \mu_X\, \mu_Y = 0, \tag{4}$$

so the covariance is zero. However, if the variables are correlated in some way, then their covariance will be nonzero. In fact, if $\operatorname{cov}(X, Y) > 0$, then Y tends to increase as X increases, and if $\operatorname{cov}(X, Y) < 0$, then Y tends to decrease as X increases. Note that while statistically independent variables are always uncorrelated, the converse is not necessarily true.

In the special case of $Y = X$,

$$\operatorname{cov}(X, X) = \left\langle X^2 \right\rangle - \langle X \rangle^2 \tag{5}$$

$$= \sigma_X^2, \tag{6}$$

so the covariance reduces to the usual variance $\sigma_X^2 = \operatorname{var}(X)$. This motivates the use of the symbol $\sigma_{X\,Y} = \operatorname{cov}(X, Y)$, which then provides a consistent way of denoting the variance as $\sigma_{X\,X} = \sigma_X^2$, where σ_X is the standard deviation.

The derived quantity

$$\operatorname{cor}(X, Y) = \frac{\operatorname{cov}(X, Y)}{\sigma_X\, \sigma_Y} \tag{7}$$

$$= \frac{\sigma_{X\,Y}}{\sqrt{\sigma_{X\,X}\, \sigma_{Y\,Y}}}, \tag{8}$$

is called statistical correlation of X and Y.

The covariance is especially useful when looking at the variance of the sum of two random variates, since

$$\operatorname{var}(X+Y) = \operatorname{var}(X) + \operatorname{var}(Y) + 2\operatorname{cov}(X, Y). \tag{9}$$

The covariance is symmetric by definition since

$$\operatorname{cov}(X, Y) = \operatorname{cov}(Y, X). \tag{10}$$

Given n random variates denoted $X_1, \ldots, X_n$, the covariance $\sigma_{ij} \equiv \operatorname{cov}(X_i, X_j)$ of X_i and X_j is defined by

$$\operatorname{cov}(X_i, X_j) = \langle (X_i - \mu_i)(X_j - \mu_j) \rangle \tag{11}$$

$$= \langle X_i X_j \rangle - \mu_i \mu_j, \tag{12}$$

where $\mu_i = \langle X_i \rangle$ and $\mu_j = \langle X_j \rangle$ are the means of X_i and X_j, respectively. The matrix (V_{ij}) of the quantities $V_{ij} = \operatorname{cov}(X_i, X_j)$ is called the covariance matrix.

The covariance obeys the identities

$$\operatorname{cov}(X+Z, Y) = \langle (X+Z)Y \rangle - \langle X+Z \rangle \langle Y \rangle \tag{13}$$

$$= \langle XY \rangle + \langle ZY \rangle - (\langle X \rangle + \langle Z \rangle)\langle Y \rangle \tag{14}$$

$$= \langle XY \rangle - \langle X \rangle \langle Y \rangle + \langle ZY \rangle - \langle Z \rangle \langle Y \rangle \tag{15}$$

$$= \operatorname{cov}(X, Y) + \operatorname{cov}(Z, Y). \tag{16}$$

By induction, it therefore follows that

$$\operatorname{cov}\left(\sum_{i=1}^{n} X_i, Y\right) = \sum_{i=1}^{n} \operatorname{cov}(X_i, Y) \tag{17}$$

$$\operatorname{cov}\left(\sum_{i=1}^{n} X_i, \sum_{j=1}^{m} Y_j\right) = \sum_{i=1}^{n} \operatorname{cov}\left(X_i, \sum_{j=1}^{m} Y_j\right) \tag{18}$$

$$= \sum_{i=1}^{n} \operatorname{cov}\left(\sum_{j=1}^{m} Y_j, X_i\right) \tag{19}$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{m} \operatorname{cov}(Y_j, X_i) \tag{20}$$

$$= \sum_{i=1}^{n} \sum_{j=1}^{m} \operatorname{cov}(X_i, Y_j). \tag{21}$$

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Bivariate Normal Distribution, Correlation Coefficient, Covariance Matrix, Statistical Correlation, Variance

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WolframAlpha covariance (3,4,5,6,8) (1,1,1,1,1) =

More things to try: = covariance (3,4,5,6,8) (1,1,1,1,1) = AAPL WMT CSCO covariance matrix return

= covariance (165; 171; 168; 165; 176; 150; 164; 181; 172; 168; 175)

REFERENCES

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Spiegel, M. R. *Theory and Problems of Probability and Statistics, 2nd ed.* New York: McGraw-Hill, p. 298, 1992.

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